Multi-Class Text Classification of E-commerce Products

Purpose of The Project

In the context of Information Access and Search Engines, e-commerce platforms are a prime example of domain-specific search. The ability to accurately retrieve relevant products depends heavily on the quality of the underlying product data. A major challenge is the automatic categorization of products based on their text descriptions.

Dataset and Methodology

The project will use the "E-commerce Text Classification" dataset from Kaggle, which contains 50,424 product descriptions. We propose to build and evaluate a system for automatically classifying e-commerce products into one of four categories ("Electronics", "Household", "Books", "Clothing & Accessories") based solely on their text descriptions.

For this classification task, three supervised learning models will be applied: Logistic Regression, Support Vector Machines, and Random Forest.

The methodology will follow a standard NLP pipeline. First, the raw product descriptions will be pre-processed. These cleaned texts will then be converted into a numerical feature space using the TF-IDF vectorization technique. Finally, the three selected models will be trained on this data and their performance will be evaluated on a held-out test set using Accuracy, Precision, Recall, and F1-Score, complemented by a Confusion Matrix to analyze misclassification patterns.

To implement this project, we will use several Python libraries. Pandas will be utilized for loading and managing the dataset. The NLTK libraries will be essential for the text pre-processing steps. Finally, Scikit-learn will be the core library for TF-IDF, implementing the supervised learning models, and calculating all evaluation metrics.

Additionally, a basic user interface will be developed to demonstrate the model's predictions on new product descriptions.

Timeline

Week 9: Data cleaning, and implement TF-IDF vectorization technique.

Week 10: Implement supervised learning models.

Week 11: Train and evaluate models. Begin performance comparison.

Week 12: Implement user interface and final checks.

Dataset

https://www.kaggle.com/datasets/saurabhshahane/ecommerce-text-classification/data

21011021 – İsmail İPEK

21011022 – Tarık TAŞKIN